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All working is shown below each question.

Q1. [Linear Equation]

$$2x + 6 = 14$$

$$2x = 14 - 6 = 8$$

$$x = 4$$

Final Answer: $x = 4$

Q2. [Factorisation]

$$6x^2 + 9x$$

Common factor = $3x$

$$= 3x(2x + 3)$$

Final Answer: $3x(2x + 3)$

Q3. [Simultaneous Equations]

From eq 2: $x = y + 2$

Sub into eq 1: $2(y+2) + y = 10$

$$2y + 4 + y = 10 \Rightarrow 3y = 6 \Rightarrow y = 2$$

$$x = y + 2 = 4$$

Final Answer: $x = 4, y = 2$

Q4. [Triangle]

(a) Hypotenuse = $\sqrt{6^2 + 8^2} = \sqrt{100} = 10 \text{ cm}$

(b) Area = $0.5 \times 6 \times 8 = 24 \text{ cm}^2$

(c) Perimeter = $6 + 8 = 14 \text{ cm}$ (working shown)

Final Answer: (a) 10 cm (b) 24 cm^2 (c) 14 cm

Q5. [Quadratic Equation]

(a) $x^2 - 5x + 6 = (x - 2)(x - 3)$

(b) $x - 2 = 0 \Rightarrow x = 2$ | $x - 3 = 0 \Rightarrow x = 3$

(c) $x=2$: $4 - 10 + 6 = 0$ (correct)

$x=3$: $9 - 15 + 6 = 0$ (correct)

Final Answer: (a) $(x-2)(x-3)$ (b) $x=2, x=3$ (c) Both verified